

Project Title

MedAssist AI – Smart Healthcare & Diagnostic System

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Milestone: Milestone 4

Role:

Full-Stack Development – Dynamic Content Routing System, Structured Public & Dashboard Page Architecture, and Real-Time Search-to-Content Integration for the MedAssist AI Web Platform.

1. Introduction

Milestone 4 focused on building a dynamic, search-connected client experience on top of the machine learning diagnostic pipeline delivered in Milestone 3. The work centered on replacing static content pages with an intelligent dynamic routing system, and on completing the full page architecture of the platform — five core public-facing pages and five core interactive dashboard pages — giving users a structured, personalized view of any medical condition, medication, or diagnostic result they search for.

2. Work Completed

2.1 Intelligent Dynamic Routing (src/app/(public)/[category]/[title]/page.tsx)

Replaced static pages with a Next.js dynamic catch-all route that parses path parameters (e.g., /diseases/viral-fever or /medications/paracetamol) to identify and fetch the correct record, so any searched disease, medication, or medical term automatically resolves to its own structured page without a manually built route.

2.2 On-Demand Data Rendering

Integrated the routing layer directly with the high-performance in-memory caching backend (/api/search), fetching the selected item's CSV row dynamically and populating the details layout in under 10 milliseconds.

2.3 Structured Grid UI Layout

Parsed CSV attributes are rendered into clean, responsive detail categories:

- Symptom Profile — matching and associated symptoms
- Severity Level — visual gauge/indicator for low, moderate, or high urgency
- Precautions & Care Steps — checklist cards for first-aid and preventative recommendations

- Specialist Advice — recommended medical specialist for the condition (e.g., Cardiologist for chest pain, Pulmonologist for breathing trouble)

2.4 Core Public Pages

- Home (/) — hero banner, real-time fuzzy-search bar resolving in under 10ms, interactive product tour, animated key statistics
- Features (/features) — 200-tree Random Forest model details (60,000 samples, 658 disease classes), Neural Brain Memory session recall, PDF/scan OCR parser
- How It Works (/how-it-works) — 4-step scroll-animated timeline (Enter Symptoms → Upload Scans/Reports → AI Processing → Get Detailed Diagnosis) with an FAQ accordion
- About Us (/about) — mission and vision, leadership and Medical Advisory Board, platform metric badges (99% uptime, 24/7 AI availability)
- Contact (/contact) — validated glassmorphic contact form with support details and response-time guarantees

2.5 Core Interactive Dashboard Pages

- Dashboard Home (/dashboard) — quick-stat cards, health trend chart, recent activity feed
- AI Symptom Checker (/dashboard/symptom-checker) — dual-mode switcher (All-in-One form / 4-step guided wizard) with a direct interface to the FastAPI ML prediction backend
- Medication Tracker (/dashboard/medications) — add-medication modal, pill-compliance calendar checklist, drug-interaction alerts
- Health Insights & Analysis (/dashboard/insights) — historical blood pressure, blood sugar, and weight trend graphs with automated LLM-generated wellness suggestions
- Medical Reports & History (/dashboard/reports) — OCR scan upload and downloadable structured clinical PDF summaries of prior diagnoses

3. Technology Stack

Layer	Technology Used
Frontend Framework	Next.js 16 (App Router, Turbopack), React 19, TypeScript
Styling & Design System	TailwindCSS v4, Framer Motion
Routing	Next.js dynamic catch-all routes
Backend Search Layer	Node.js in-memory CSV search (/api/search)
Backend Language	Python 3.11
ML Microservice Framework	FastAPI
Authentication & Database	Firebase Auth / Firestore

LLM Engine	Groq API (Llama 3.3 70B)
Document Generation	ReportLab, jsPDF
Containerization	Docker

4. Verification & Performance Status

All routing, search, and page-rendering components were verified with the following results:

- Dynamic Routing: verified across both /diseases/ and /medications/ category paths, correctly resolving to structured detail pages
- Search-to-Page Resolution: consistently resolves and renders in under 10 milliseconds
- Public Pages: all 5 core public pages verified for responsiveness and content accuracy
- Dashboard Pages: all 5 core interactive dashboard pages verified for responsiveness and functional interactivity
- Build & Compilation: all source code and page routes verified, committed, and tested cleanly.

5. Summary of Accomplishments

Feature	Milestone 3 State	Milestone 4 Complete State
Content Routing	Static pages	Dynamic catch-all routing ([category]/[title])
Data Rendering	N/A	On-demand CSV row rendering via /api/search (<10ms)
Detail Page Layout	N/A	Structured grid: Symptom Profile, Severity, Precautions, Specialist Advice
Public-Facing Pages	Not defined	5 core pages: Home, Features, How It Works, About Us, Contact
Dashboard Pages	Symptom Checker only	5 core pages: Dashboard Home, Symptom Checker, Medication Tracker, Insights, Reports

6. Conclusion

During this milestone, MedAssist AI gained an intelligent dynamic routing system that resolves any searched disease or medication to a structured, on-demand detail page in under 10 milliseconds, along with a completed page architecture spanning five core public-facing pages and five core interactive dashboard pages. Together, these deliverables move the platform from a diagnostic-only interface toward a fully navigable, search-connected healthcare product.